

AI Interview Trainer Agent

Project Plan for AICTE–IBM SkillsBuild Problem Statement No. 22

1. Problem Statement

An Interview Trainer Agent, powered by Retrieval-Augmented Generation (RAG), prepares users for job interviews by generating tailored question sets and preparation strategies based on profile, experience level, and job role. It retrieves role-specific interview questions, industry expectations, behavioral scenarios, and HR guidelines. Users can provide a resume or job title and receive targeted questions, model answers, and improvement tips. The solution supports both technical and soft-skill assessment. IBM Cloud Lite services / IBM Granite is mandatory.

2. Proposed Solution

AI Interview Trainer — Your Personal AI Interview Coach

The system acts as an intelligent interviewer that analyzes a candidate's resume/profile, identifies the target role and experience level, retrieves relevant interview knowledge through RAG, conducts a personalized mock interview, evaluates answers, asks adaptive follow-up questions, and generates a final performance report.

3. Core Workflow

Step	Module	Function
1	Candidate Profile	Collect name, target role, experience, skills and resume.
2	Resume Analyzer	Extract skills, education, projects, experience, certifications and technologies.
3	RAG Knowledge Base	Retrieve relevant technical, HR, behavioral and situational interview material.
4	IBM Granite	Generate grounded, role-specific interview questions and responses.
5	Mock Interview	Conduct an interactive interview and generate contextual follow-up questions.
6	Answer Evaluation	Assess technical accuracy, communication, relevance and answer quality.
7	Final Report	Provide scores, strengths, weaknesses and a personalized preparation roadmap.

4. Agentic AI Architecture

Candidate → Profile Analyzer → Role/Experience Analysis → RAG Knowledge Base → IBM Granite → Interview Session → Answer Evaluation → Adaptive Follow-up → Performance Report

5. RAG Knowledge Base

- **Technical:** role-specific questions, concepts, coding and system fundamentals.
- **HR:** common HR questions, expectations and interview guidance.
- **Behavioral:** teamwork, leadership, conflict and workplace scenarios.
- **Situational:** role-specific real-world situations and decision-making questions.
- **Industry expectations:** preparation guidance relevant to the selected job role.

6. Suggested MVP Features

- Resume PDF upload and candidate profile creation.
- Target job-role and experience selection.
- RAG-powered personalized question generation.
- Interactive mock interview with technical and HR questions.
- Dynamic follow-up questions based on the candidate's previous answer.

- Answer evaluation with score and actionable feedback.
- Final performance dashboard and preparation roadmap.

7. Suggested Technology Stack

Layer	Suggested Technology
Frontend	React / Next.js
Backend	Node.js + Express
AI Model	IBM Granite
Cloud	IBM Cloud Lite services
RAG	Document ingestion + chunking + embeddings + vector retrieval
Database	Lightweight database for users and interview sessions

8. Example Live Demo

Stage	Example
Candidate	Software Developer — Fresher
Skills	Python, JavaScript, React, Node.js, SQL, REST APIs
Question	Explain the difference between let, const and var in JavaScript.
Candidate Answer	User provides a natural-language answer.
Evaluation	AI checks correctness, clarity, relevance and completeness.
Follow-up	AI asks a deeper question based on the previous response.
Final Output	Technical, communication, problem-solving and HR scores + improvement plan.

9. Why This Is Agentic AI

The solution is more than a static chatbot. It analyzes candidate context, retrieves relevant knowledge, selects and generates questions, evaluates responses, adapts the next question to performance, and produces a personalized preparation plan. This demonstrates a multi-step autonomous workflow grounded in retrieved information.

10. Implementation Strategy

Phase	Goal
Phase 1	Build working prototype: profile → questions → answers → feedback.
Phase 2	Add RAG knowledge base and grounded retrieval.
Phase 3	Integrate IBM Granite for generation and evaluation.
Phase 4	Add adaptive follow-up questioning and final dashboard.

11. Judge-Facing One-Line Pitch

“An AI-powered personal interview coach that understands your profile, retrieves role-specific knowledge, conducts an adaptive mock interview, evaluates your answers, and tells you exactly how to improve.”

Source basis: AICTE–2026 IBM Skills Build for University Engagements problem statement document, Problem Statement No. 22, Interview Trainer Agent.